Varun Patil

Security & Distributed Systems Researcher

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Profile

Research leader with expertise in secure networking, trusted compute, distributed synchronization, and large-scale infrastructure. End-to-end engineer with proven record of implementing and productionizing security-sensitive products at Internet scale, building across all layers of the stack.

Professional Experience

Meta Platforms, Inc. — Menlo Park, California
Research Scientist (E5) – WhatsApp — *Jun. 2025 – present*

- Led the development of the secure and transparent data plane of Meta's Private AI Inference platform
  - Productionized **WhatsApp's Incognito AI**, an **industry-first product** enabling truly private AI interactions
  - Designed the end-to-end encryption protocol for the Trusted Execution Environment (TEE) platform
  - Led the cross-team effort to securely scale the TEE platform for orchestrating complex AI services
  - Implemented the Remote Attestation library serving as the central trust and security component
  - Optimized the data plane across 4 stack layers, improving end-to-end P50 latency by 60%
- Served as the production **incident response expert** for all WhatsApp services running on the Private AI platform
- Built manual and automated review frameworks for enforcing data handling practices for sensitive services

Operant Networks — Los Angeles, California
Software Developer (C++) – Security — *Jun. 2023 – Jun. 2025*

- Implemented a cross-platform TPM-backed key storage solution for the Defined-Trust Transport (DeftT)
- Designed a new testing framework, internal tooling based on Conan, and 3x faster cross-platform CI/CD pipelines
- Designed a retrofit field device remote access platform with flexible ACL controls

University of California, Los Angeles — Los Angeles, California
Graduate Student Researcher – Internet Research Laboratory — *Apr. 2021 – Jun. 2025*

- Drove bleeding-edge research on Named Data Networking (NDN), a future Internet protocol stack
  - [1] Patil, Desai, Zhang, 2022. *Kua: A Distributed Object Store over Named Data Networking*. ACM ICN '22
  - [2] Moll, Patil, Zhang, Pesavento, 2021. *Resilient Brokerless Publish-Subscribe Over NDN*. IEEE MILCOM '21
- Designed the State Vector Sync transport protocol for distributed synchronization in adversarial networks
- Implemented NDN Distance Vector (ndn-dv), an efficient multi-path routing protocol for named data networks

Open Source

Memories | PHP + Vue | 3,800+ GitHub stars

- Efficent, beautiful and feature-rich photo management app for Nextcloud, supporting very large media libraries
- Project owner and lead developer, collaborating with a large user community and 40+ developers globally

Named Data Networking Daemon | Golang

- High performance data-centric networking stack implementation in user space including a packet forwarder

Education

University of California, Los Angeles — California, USA
Ph.D. Computer Science (Advisor: Lixia Zhang) — *Dec. 2020 – Jun. 2025*
M.S. Computer Science — *Sep. 2020 – Dec. 2021*

Indian Institute of Technology Bombay — Mumbai, India
B.Tech. Mechanical Engineering — *Jun. 2016 – Aug. 2020*
Minor in Computer Science and Engineering, Institute Technical Roll of Honour Award